

Phase 1 Task Breakdown: Cross-Device AI Assistant

Every task needed to produce the Phase 1 report. No code is delivered in this phase. Pick the tasks you want to own and write your name in the Owner column.

The system we are designing

One AI assistant runs locally on the user's strongest device. It connects their PC, laptop, phone and iPad peer-to-peer over the local network, with no internet and no cloud server, and understands files and data across all of them.

Features: duplicate detection · unified search · password management · clipboard sync · screen sharing with remote control · extra screen · direct file sharing · offline device records.

What Phase 1 must deliver (from the supervisor's sample reports)

1. **Introduction:** overview, scope, objectives, motivation, plan
 2. **Literature Review:** background, related systems, contribution
 3. **Problem Analysis:** problem spec, functional & non-functional requirements, evaluation plan
 4. **System Design:** architecture, DFD (Level 0 & 1), sequence diagrams, use case diagram
- Plus abstract, references, lists of tables and figures.

Size: S 1 point, a few hours M 2 points, 2–4 days L 3 points, about a week · Total: 67 points, about 13–14 per member

A. Report writing Chapters 1–3 and front/back matter

ID	TASK	WHAT IT INCLUDES	SIZE	OWNER
A1	Chapter 1: Introduction	Introduction, project overview, scope (in and out), aim and objectives, motivation, report outline, summary.	M	-----
A2	Project plan & schedule	Task schedule tables for weeks 1–6, 7–12 and 12–17 (task, start, end, status), kept up to date.	S	-----
A3	Abstract	One-page abstract, written last from the finished chapters.	S	-----
A4	Chapter 2: Background	Concepts the system relies on: P2P networking and local discovery, on-device AI/LLMs, duplicate detection (hashing, perceptual hashing, embeddings), password vault encryption, screen streaming.	M	-----
A5	Chapter 2: Existing related systems	Review similar products and papers (e.g. Syncthing, KDE Connect, LocalSend, Apple Continuity/Sidecar, Windows Phone Link, Bitwarden/KeePass, Spacedesk/Duet, dupeGuru/Czkawka) with a comparison table.	L	-----
A6	Chapter 2: Contribution	The gap our system fills: one local AI assistant, all features, no account or cloud.	S	-----
A7	Chapter 3: Problem specification	Clear statement of the problem and who it affects.	S	-----
A8	Chapter 3: Functional requirements	Compile and number all functional requirements. Each feature owner (section C) drafts the requirements for their own feature. This task merges them and keeps them consistent.	S	-----
A9	Chapter 3: Non-functional requirements	Privacy, security, performance, offline operation, battery and resource use, cross-platform support, usability.	M	-----

A10	Implementation & evaluation plan	How each feature will be built and tested in Phase 2: test types, success criteria, target devices.	M	-----
A11	References & final assembly	IEEE references, table of contents, lists of tables and figures, consistent formatting, final proofreading.	M	-----

B. Overall system design Chapter 4: the whole system

ID	TASK	WHAT IT INCLUDES	SIZE	OWNER
B1	System architecture	Layered architecture diagram and description: device agent, P2P layer, AI node, feature modules, local index/database, user interface.	L	-----
B2	DFD Level 0 (context)	Context diagram, plus the external entities table.	M	-----
B3	DFD Level 1	Main internal processes and data flows, plus the processes table.	L	-----
B4	Use case diagram	All actors and use cases, with a short description of each use case.	M	-----
B5	Data model	Entities and relationships: devices, file index, hashes, duplicate groups, vault entries, sessions, sharing history.	M	-----
B6	Technology & platform choices	Languages and frameworks, local AI runtime and model, P2P library, and a platform support matrix (Windows / macOS / Android / iOS).	M	-----

C. Feature design For each feature: how it works, a sequence diagram, key design decisions, and its functional requirements

ID	TASK	WHAT IT INCLUDES	SIZE	OWNER
C1	Device discovery, pairing & trust	How devices find each other on the local network, how a new device joins (e.g. QR code pairing), and how devices trust each other. Every other feature depends on this.	L	-----
C2	AI assistant node	Choosing the strongest device, which local model to run, what the assistant does (e.g. natural-language requests like "find my CV"), and what happens when that device goes offline.	L	-----
C3	File indexing & offline records	How each device indexes its files, how indexes are shared, and how the last-known state of an offline device is kept and shown.	M	-----
C4	Duplicate file detection	Exact duplicates (hashing), similar files (photos and videos via AI), and how results are presented. <i>Choosing which device keeps the file: to be confirmed.</i>	L	-----
C5	Unified search	Searching all devices from any device, including offline devices through their records.	M	-----
C6	Direct file sharing	Sending files between devices over the local network: transfer flow, large files, interruptions.	M	-----
C7	Password management	Collecting accounts and passwords per device, the encrypted vault, syncing it between devices, and the threat model.	L	-----
C8	Clipboard sync	Copy on one device and paste on another, for text and images.	S	-----
C9	Screen sharing & remote control	Streaming a device's screen, sending mouse and keyboard input back, and the permissions each platform requires.	L	-----
C10	Extra screen	Using the iPad or another device as a second monitor, including a feasibility study of platform limits (especially iOS).	L	-----
C11	Security & privacy design	Encryption between devices, access control, what data is stored where, protection if a device is lost.	M	-----

D. User interface design

ID	TASK	WHAT IT INCLUDES	SIZE	OWNER
D1	Screen wireframes	Main screens for desktop and mobile: device dashboard, duplicates, search, password vault, sharing, screen tools.	L	-----
D2	User flows	Step-by-step journeys: first setup and pairing, cleaning duplicates, finding a file, sending a file.	M	-----

E. Coordination

ID	TASK	WHAT IT INCLUDES	SIZE	OWNER
E1	Team & supervisor coordination	Meetings with the supervisor, meeting notes, tracking progress against the schedule (A2).	S	-----
E2	Phase 1 presentation	Slides for the Phase 1 discussion, built from the report.	M	-----

Optional: five balanced tracks A starting point if the team prefers one track per member. Swap tasks freely.

1 • Architecture & Connectivity
14 points

- B1 Architecture
- B2 DFD Level 0
- B6 Technology choices
- C1 Discovery & pairing
- C8 Clipboard sync
- C11 Security design
- E1 Coordination

2 • AI & Files
13 points

- C2 AI assistant node
- C3 Indexing & offline records
- C4 Duplicate detection
- C5 Unified search
- B5 Data model
- A8 Functional requirements

3 • Security & Sharing
14 points

- C7 Password management
- C6 File sharing
- B3 DFD Level 1
- B4 Use case diagram
- A9 Non-functional requirements
- A6 Contribution
- A7 Problem specification

4 • Screens & Interface
13 points

- C9 Screen sharing
- C10 Extra screen
- D1 Wireframes
- D2 User flows
- E2 Presentation

5 • Research & Report
13 points

- A1 Introduction
- A2 Plan & schedule
- A3 Abstract
- A4 Background
- A5 Related systems
- A10 Evaluation plan
- A11 References & assembly